Algorithmic optimization for maximizing scale factor control precision with an all–digital modulator architecture for an iFOG

Sławomir Niespodziany

***Abstract*— This paper presents the design and optimization of a modulator subsystem for closed-loop interferometric fibre optic gyroscopes, with a focus on maximizing scale factor control precision through an all-digital hardware architecture. The operating principle of the gyroscope is reviewed, followed by a description of the four-step modulation scheme used to linearize the optical response, maintain maximum sensitivity, and enable rotation direction detection. Two modulator architectures are analyzed and compared: a mixed analog-digital design employing a dedicated programmable gain stage, and a fully software-defined all-digital implementation. The all-digital approach is advocated for its reduced component count, fewer nonlinearity sources, and greater flexibility during development. A mathematical analysis of quantization error is derived to characterize the achievable scale factor control accuracy as a function of digital-to-analog converter resolution and reference voltage selection. The derivation reveals that the scale factor feedback signal depends on four consecutive drive samples rather than two, a non-obvious result with direct consequences for algorithm correctness. Based on this analysis, common implementation pitfalls are identified and discussed, including insufficient sample history, drive asymmetry due to discrete amplitude representation, and resolution loss caused by symmetric rounding of signed values. An optimal algorithm is presented, utilizing fixed-point arithmetic with concrete rounding method to recover one bit of effective resolution relative to naive implementations. The results establish a software foundation for high-precision closed-loop interferometric fibre optic gyroscope modulator design.**

***Index Terms*— all-digital modulator, fibre optic gyroscope, fixed-point arithmetic, four-step modulation, scale factor control, Sagnac effect.**

I. Introduction

iFOG (Interferometric Fibre Optic Gyroscope) concept is based on the Sagnac effect [1], discovered in 1913 [2]. Over the years, its design has been implemented and iterated multiple times, making it well refined. However, the limits which hold the development back are constantly pushed forward as technology evolves. This allows for better precision, miniaturized size, lower power consumption, etc. At the same time, the ever progressing industry finds new and more strict applications for the gyroscopes, setting the requirements bar even higher. Such a context drives the already existing designs to be tested and optimized in every possible aspect. This applies equally to all the optical and electro–optic components, to the electronics, and to the driving software. A crucial part of iFOG control system is the modulator. In a closed–loop design [3] it is a vital element of the feedback loop, directly affecting overall performance. It drives the optical circuit into a predefined state, in which the circuit's response can be demodulated and error signals determined.

This paper takes a closer look at the all–digital implementation of the modulator subsystem — in particular its electronic architecture and the algorithm controlling it. Mathematical analysis is conducted to allow for pure–software optimizations and an exemplary implementation is proposed to fully utilize the offered capabilities.

II. Principle of operation - open–loop configuration

The light source of an iFOG generates a beam, which is ideally split in a 50/50 ratio. These halves enter a loop of a relatively long fiber through each of its ends. The fiber length L determines the light transit time τ through it (1), where n is the refraction coefficient, affecting light speed in the medium.

$$\tau = \frac{L}{n \cdot c} \tag{1}$$

The halves go past each other and exit on the opposite ends of the fiber. The optical circuit is arranged, so that the two exiting beams interfere with each other and the resulting light intensity is measured at a detector. When the circuit stays at rest, the beams interfere with zero phase shift ($\Delta\phi = 0$), resulting in maximum intensity P_0. However, when the device rotates, the two halves travel paths of a slightly different length, resulting in non–zero phase shift. The relative shift at the interference point is rotation dependent and in this paper it is referred to as the Sagnac–induced phase shift

$$\Delta\phi_s = \frac{2\pi \cdot L \cdot D \cdot n}{\lambda \cdot c} \cdot \omega \tag{2}$$

where D - fiber loop diameter, ω - angular velocity, λ - wavelength.

This has been depicted in Figure 1, where both halves enter the loop at the *Entry point* and after the transit time exit through the same point - now referred to as the *Exit point*. Because of the non–zero rotation, this point moves along the loop, effectively shrinking one path and extending the other. Path difference is relatively small, thus it is expressed as a phase difference in wavelength domain.

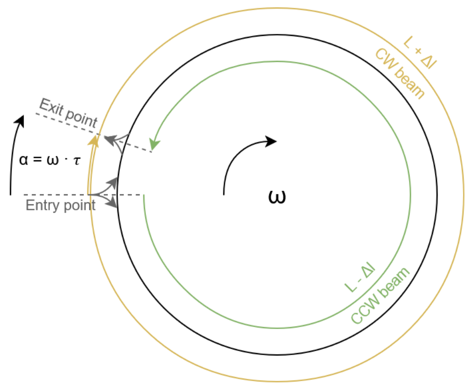


Fig. 1: Two counterpropagating light beam halves enter the fiber loop at the *Entry point*. Their phase is initially in sync. After transit time τ they exit and interfere. During that time the *Entry/Exit point* moves by the angle $\alpha = \omega \cdot \tau$ according to the rotation rate ω. As a result the CW and CCW propagating halves cover non–equal path lengths ($L \pm \Delta l$), which leads to a non–zero phase difference.

Equation (3), depicted in Figure 2, is the optical circuit response function [4]. It describes how the measured light intensity depends on the total phase shift of the interfering light.

$$P(\Delta\phi) = \frac{P_0}{2} \cdot (1 + cos(\Delta\phi)) \tag{3}$$

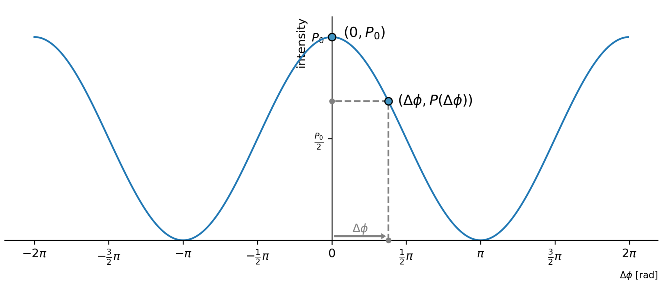


Fig. 2: Optical circuit response is an upshifted cosine. The two exemplary operating points refer to the circuit being at rest, yielding the maximum possible intensity in $(0, P_0)$ and a circuit subjected to rotation, at non–zero phase shift and decreased intensity in $(\Delta\phi, P(\Delta\phi))$.

When the circuit is at rest ($\omega = 0 \rightarrow \Delta\phi = 0$), the operating point resides at peak intensity P_0. Non–zero rotation results in phase shift which moves the operating point according to rotation rate and its direction, decreasing the measured intensity. Two problems are linked with this *open–loop* operation [5]. Firstly, intensity drop alone does not allow for determining rotation direction — the response function is symmetrical. Secondly, in the resting region, the slope is flat, thus the sensitivity is zero. With greater rotation rate, the slope becomes more steep, reaching maximum sensitivity at $\Delta\phi = \pm\frac{\pi}{2}$, but then decreases again. Both issues are resolved with *closed–loop* operation described in the following section.

III. Modulation scheme - closed–loop operation

In *closed–loop* operation [5] the control system biases the optical circuit sequentially in four predefined points $\Delta\phi \in \{+\frac{3}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi, +\frac{1}{2}\pi\}$, utilizing *four–step* modulation [6], depicted in Figure 3. This allows for linearizing the response, keeping the circuit at maximum sensitivity and detecting rotation direction.

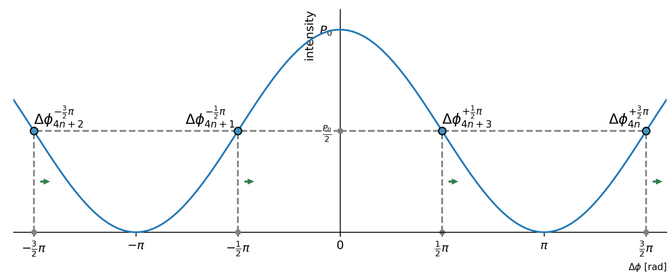
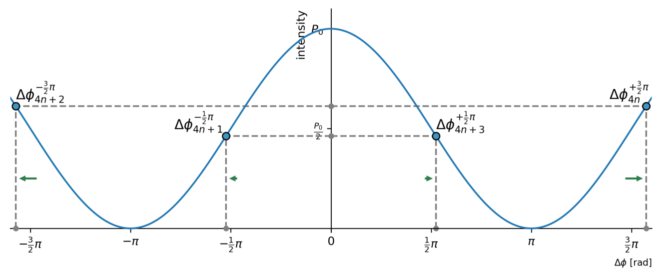


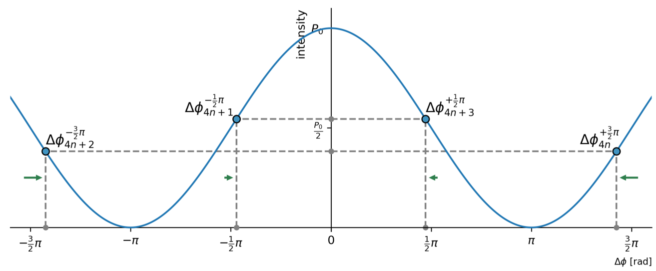
Fig. 3: Circuit biased in four points of highest sensitivity.

Compared to *two–step* modulation [6], this scheme is superior, as it allows for controlling the *scale factor* - an essential parameter for maintaining device performance. The modulator injects the mentioned phase shifts by using an electro–optic actuator placed on the fiber path. The amount of applied shift is controlled by an electronic circuit. It performs digital calculations and generates corresponding voltages via a digital–to–analog converter and an electronic front–end. The voltage is linearly translated to phase shift according to the half–wave voltage V_π - a parameter of the actuator. Each element in this pipeline is subject to temperature drift and aging effects [7], which influence the total combined gain. For this reason, continuous calibration is required for maintaining measurement accuracy. It is realized by tracking the scale factor during operation. This allows the control algorithm to know where the desired bias points are in the digital domain, while they are allowed to constantly drift away.

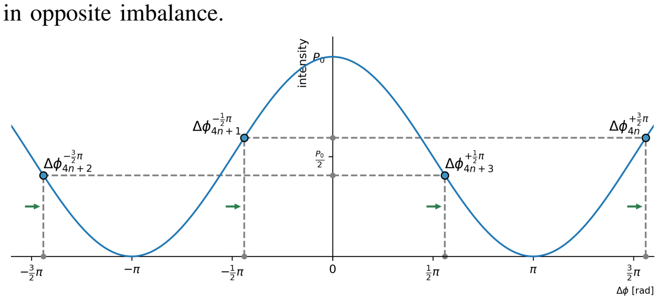
Diagrams in Figure 4 depict how the biased circuit responds to various conditions - i.e. misaligned scale factor and uncompensated rotation present. While this paper does not discuss in detail how the rotation is measured, it is crucial to emphasize that different feedback signals can be derived for scale factor adjustment and rotation compensation. These feedback signals allow the digital control loop to adjust runtime parameters and place the operating points exactly where they are intended to be.



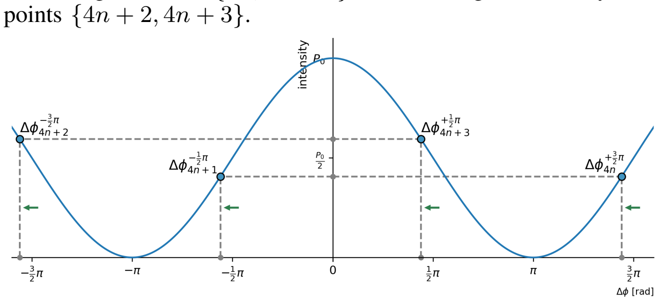
(a) Scale factor too high - bias points are shifted away from zero, resulting in intensity imbalance.



(b) Scale factor too low - bias points shifted towards zero, resulting in opposite imbalance.



(c) Phase shift induced by rotation in one direction - bias points shifted right. Points $\{4n, 4n+1\}$ result in higher intensity than points $\{4n+2, 4n+3\}$.



(d) Phase shift induced by rotation in opposite direction - bias points shifted left. Points $\{4n, 4n+1\}$ result in lower intensity than points $\{4n+2, 4n+3\}$.

Fig. 4: Four–step modulation allows for detection of scale factor error and uncompensated Sagnac–induced phase shift.

From a control point of view and for simplicity, the total combined scale factor of the processing pipeline can be referred to as the half–wave voltage V_π. Even though the actuator gain drift itself is only one of the scale factor components, the assumed linearity of others makes it convenient to aggregate all into a single parameter. This paper assumes them to be included in V_π. Other non–linear errors, such as DAC INL, are out of scope of this analysis.

Control system uses the combined scale factor to bias the optical circuit continuously with the following bias point order

$$\begin{aligned}\{\Delta\phi_{4n+0}^{+\frac{3}{2}\pi} &\rightarrow (+\frac{3}{4}\pi),\\ \Delta\phi_{4n+1}^{-\frac{1}{2}\pi} &\rightarrow (-\frac{1}{4}\pi),\\ \Delta\phi_{4n+2}^{-\frac{3}{2}\pi} &\rightarrow (-\frac{3}{4}\pi),\\ \Delta\phi_{4n+3}^{+\frac{1}{2}\pi} &\rightarrow (+\frac{1}{4}\pi)\}\end{aligned} \tag{4}$$

where the superscripted phase is the intended effective phase shift for the given sample and the value on the right is the amount of shift required to reach that intended value. The bias points (or samples) are sequentially numbered.

The modulator runs with a period of half the transit time - $\frac{\tau}{2}$. This allows for the effective phase shift to be a result of a relative difference between every k'th and $(k-2)$'th samples. In such case the phase shift applied by the modulator for each sample can be half of the desired amplitude, reducing the overall required voltage range [5]. Additionally to this sequence, to null the effect of Sagnac–induced phase shift, modulator adds an ever–updating ramp signal, utilizing Serrodyne modulation [3] with the increment value of $\Delta\phi_r = -\Delta\phi_s$ - a phase shift of equal amplitude, but opposite sign. This way the algorithm nulls out the Sagnac–induced shift. While knowing how much of it has been cancelled, it knows the exact rotation rate. In every cycle, the software monitors both feedback signals and adjusts control accordingly.

IV. Modulator hardware architecture

This paper advocates for an all–digital modulator architecture. Three distinct approaches can be distinguished:

(i) fully analog,
(ii) mixed (digital with a dedicated, analog gain stage),
(iii) all–digital.

Firstly - a fully analog variant (i), while feasible, does not comply with today's standards, where information is processed in a digital form. For integration with external systems, an analog–to–digital interface needs to be present anyway. In such case, incorporating an MCU or DSP in the processing loop brings the advantage of performing certain operations on pure data, rather than on analog signals carrying these data. A modern solution shall consist of a digital part for performing mathematical operations and an electronic part for interfacing with the optical circuit - this resembles a classical mixed implementation (ii). It consists of a DAC and a dedicated gain stage. Lastly, in a world of *software–defined* solutions [8], [9], an architecture with the minimum possible amount of electronic components seems to be sound. The two latter concepts are described in this section. The all–analog architecture will not be discussed.

A. Dedicated gain stage - mixed variant

A reasonable tradeoff between analog and digital approaches is the architecture proposed in [5]. It is composed of a digital part, a DAC at the boundary and a dedicated gain stage on the analog side. It has been depicted in Figure 5.

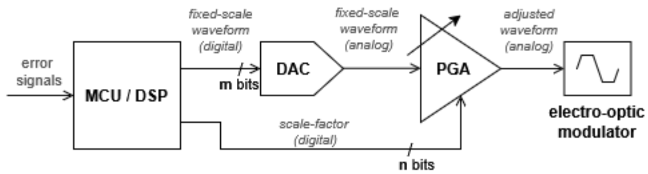


Fig. 5: Modulator with a dedicated analog gain stage. MCU/DSP outputs a fixed–scale digital waveform, which directly drives a DAC with n–bit resolution. Then, before reaching the electro–optic actuator, the scale factor is applied by a programmable gain amplifier (PGA), or a custom analog gain stage, driven with m–bit word. The resulting waveform is scale–adjusted.

In this configuration, the DAC and PGA are driven with $n-$ and m–bit words. The digital nature of controlling the gain stage introduces unavoidable quantization noise into the control loop. One of the important advantages of this setup is the intrinsic separation of the modulation problem into two smaller parts - generating the waveform in fixed–scale and applying the scale factor afterwards. The two can be implemented by software without dependence on each other, resulting in two distinct control loops. On the downside, the dedicated gain stage is subject to interference. It may have its own nonlinearities and introduce delay in the signal path. Most importantly, its bandwidth may be gain-dependent, distorting the outgoing waveform.

Since the scale factor is not expected to vary by more than a few per cent over the device's lifetime, the PGA gain range is set for the worst–case scenario and rarely uses its full span (the m–bit word). This means that even if the effective operating point were to drop to 50% of the available range — well beyond realistic — the performance would remain no worse than with $(n-1)$–bit control. This observation motivates removing the PGA entirely and applying the scale factor digitally instead, leading to the all–digital architecture described next. Selecting a DAC with one additional bit directly compensates for this loss.

B. Software–defined solution - all–digital variant

The all–digital approach depicted in Figure 6 utilizes the availability and wide range of high precision digital–to–analog converters. It enables eliminating the physical gain stage, by applying the scale factor digitally, prior to driving the DAC. This requires DAC reference voltage to be high enough to cover all possible ranges. This approach also introduces quantization noise. These two disadvantages are similarly present in the previous solution. The main advantage this concept brings is minimization of the amount of electronic components used. This yields fewer sources of potential nonlinearities and electromagnetic interference. Since the electro–optic modulator has high input impedance, the output of a voltage–DAC can be directly connected to it, omitting any buffers on the signal path. The benefit of less noisy electronics compensates the unavoidable DAC resolution loss — its full range may never be utilized because of the required margins. Using a DAC with one additional bit - relative to the solution with a dedicated gain stage - compensates for the range margins. Moreover, the following sections discuss an algorithmic approach which regains one further bit of accuracy when using the all–digital architecture.

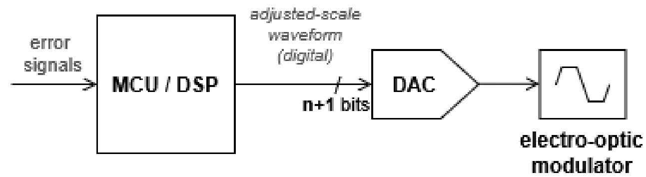


Fig. 6: All–digital modulator consists of minimal number of electronic components. In some applications this can simply be a DAC alone.

An additional advantage of the proposed architecture is that the whole design becomes a software–defined solution. As in other fields, such an approach brings flexibility from the R&D process perspective. This is mainly because, while iterating hardware is a time–consuming process, updating software is relatively quick and cost-effective. Moving the problem completely to the software side obviously makes the implementation more complex, but at the same time it limits the amount of interaction at the boundary between software and hardware development, significantly reducing overhead. From this perspective the net workload remains relatively stable, offering a slightly different — more modern — approach.

1) Voltage reference: For the all–digital modulator an adequate reference voltage must be set to account for the maximum possible scale factor in the device's lifetime. From (4) it follows that a span of 1.5π is required for biasing. Additionally, the Serrodyne modulator requires a span of 2π. This yields the total span of $S = 3.5\pi$. Minimum DAC reference V_{ref}^{min} shall account for it with some margin $M \in [1.0; 1.1]$

$$V_{\text{ref}}^{\text{min}} = V_{\pi}^{\text{max}} \cdot S \cdot M. \tag{5}$$

The V_{π}^{max} denotes the highest expected half–wave voltage. When determining the final reference voltage it is advantageous to round it up to a multiple of one of typically available chip references (6), e.g. $K \times 5V$ or $K \times 4.096V$. It makes implementing the electronic circuit more straightforward. In practice, a fair compromise is to utilize the resulting voltage excess for the margin M, so that after adding it, the $V_{ref} \approx V_{ref}^{min}$.

$$V_{\text{ref}} = \lceil V_{\text{ref}}^{\text{min}} \rceil \tag{6}$$

2) Drive accuracy and quantization error: The use of a DAC introduces quantization error. Converter resolution N and its reference voltage determine the minimal controllable voltage change as

$$\Delta v = V_{\text{ref}} \cdot 2^{-N}. \tag{7}$$

In the wavelength domain, the scale factor can thus be controlled with the accuracy $\Delta\phi_v$ corresponding to Δv, relatively to the actual scale factor:

$$\Delta\phi_v = \frac{\Delta v}{V_\pi}. \tag{8}$$

Due to the scale–factor dependence, it will slowly vary over time, but its magnitude shall remain stable, as the drift is only expected to be at the level of per cents. By substituting previous formulas, the final control accuracy $\Delta\phi_v$ and quantization error $\Delta\phi_q$ are

$$\begin{aligned} \Delta\phi_v &= \frac{\lceil V_\pi^{\max} \cdot S \cdot M \rceil \cdot 2^{-N}}{V_\pi} \\ \Delta\phi_q &= \frac{1}{2} \cdot \Delta\phi_v \end{aligned} \tag{9}$$

An exemplary calculation based on real–world values yields

$$\begin{aligned} \Delta\phi_v^{\text{ex}} &= \frac{\lceil \overbrace{4.5V}^{V_\pi^{\max}} \cdot \overbrace{3.5}^{S} \cdot \overbrace{1.025}^{M} \rceil \cdot \overbrace{2^{-16}}^{2^{-N}}}{\underbrace{3.5V}_{V_\pi}} = \\ &= \frac{\lceil 16.14V \rceil \cdot 2^{-16}}{3.5V} = \frac{\overbrace{16.384V}^{4 \cdot 4.096V} \cdot 2^{-16}}{3.5V} = \\ &= 2^{-16} \cdot 4.68 = 2^{-16} \cdot 2^{2.23} \\ \Delta\phi_q^{\text{ex}} &= \frac{1}{2} \cdot \Delta\phi_v^{\text{ex}} = 2^{-16} \cdot 2^{1.23} \end{aligned} \tag{10}$$

From this it is clear, that the quantization error, bound to the controllable resolution, can be improved by selecting a converter with greater number of bits. However, any selected resolution will always be subject to losing just over two bits of resolution. It must be noted that most of this loss is due to the span S, required for proper operation of the algorithm. It is not a consequence of using the proposed all–digital architecture. The quantization error follows the achieved accuracy. In each of the bias points, the following drive levels are possible

$$\begin{aligned} \phi_{4n+0}^{+\frac{3}{2}\pi} &= k_{4n+0} \cdot \Delta\phi_v = +\frac{3}{4}\pi + \delta_{4n+0} \\ \phi_{4n+0}^{-\frac{1}{2}\pi} &= k_{4n+1} \cdot \Delta\phi_v = -\frac{1}{4}\pi - \delta_{4n+1} \\ \phi_{4n+0}^{-\frac{3}{2}\pi} &= k_{4n+2} \cdot \Delta\phi_v = -\frac{3}{4}\pi - \delta_{4n+2} \\ \phi_{4n+0}^{+\frac{1}{2}\pi} &= k_{4n+3} \cdot \Delta\phi_v = +\frac{1}{4}\pi + \delta_{4n+3} \end{aligned} \tag{11}$$

where $k \in [-2^{N-1}, 2^{N-1} - 1)$ is an integer, in range of the used DAC (of N resolution). By incrementing/decrementing k_n the drive error δ_n can be controlled with the accuracy of $\Delta\phi_v$, but because of the quantization error its value can always reachup to $\Delta\phi_q$.

V. Precision estimation and impact

To close the scale factor control loop, a feedback signal is required. It is calculated from the difference of incident light intensity between two consecutive bias points. The total phase difference $\Delta\phi_n$, affecting intensity for the sample n, results from the current drive ϕ_n and the drive two points prior:

$$\Delta\phi_n = (\phi_n - \phi_{n-2}) + \underbrace{\Delta\phi_{s,n} + \Delta\phi_{r,n}}_{=0} \tag{12}$$

The values of $\Delta\phi_{s,n}$ and $\Delta\phi_{r,n}$ denote Sagnac–induced shift and the ramp shift nulling it and are assumed to be zero for this analysis. Using (3), (4), (11) and (12) and assuming the *cos* function linear near the bias points (14), intensity values at each point can be calculated as

$$\begin{aligned} P_{4n+0}^{+\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + cos(\Delta\phi_{4n+0}^{+\frac{3}{2}\pi})) = \\ &= \frac{P_0}{2} \cdot (1 + cos(\underbrace{\phi_{4n+0}^{+\frac{3}{2}\pi}}_{\frac{3}{4}\pi+\delta_{4n+0}} - \underbrace{\phi_{4n-2}^{-\frac{3}{2}\pi}}_{-(\frac{3}{4}\pi+\delta_{4n-2})})) = \\ &= \frac{P_0}{2} \cdot (1 + cos(\frac{3}{2}\pi + \delta_{4n+0} + \delta_{4n-2})) = \\ &= \frac{P_0}{2} \cdot (1 + (\delta_{4n+0} + \delta_{4n-2})) \\ P_{4n+1}^{-\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 - (\delta_{4n+1} + \delta_{4n-1})) \\ P_{4n+2}^{-\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + (\delta_{4n+2} + \delta_{4n+0})) \\ P_{4n+3}^{+\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 - (\delta_{4n+3} + \delta_{4n+1})) \end{aligned} \tag{13}$$

$$\begin{aligned} cos(\phi_b + \Delta\phi) &= \underbrace{cos(\phi_b)}_{=0} + cos'(\phi_b) \cdot \Delta\phi = \\ &= -sin(\phi_b) \cdot \Delta\phi \\ &\text{for } \phi_b \in \{+\frac{3}{2}\pi, +\frac{1}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi\} \end{aligned} \tag{14}$$

Scale factor feedback signal calculated at samples $4n + 1$ and $4n+3$ results from the intensity difference within each sample pair

$$\begin{aligned} e_{4n+1}^{-\frac{1}{2}\pi} &= P_{4n+1}^{-\frac{1}{2}\pi} - P_{4n+0}^{+\frac{3}{2}\pi} = \\ &= -\frac{P_0}{2} \cdot (\delta_{4n+1} + \delta_{4n+0} + \delta_{4n-1} + \delta_{4n-2}) \\ e_{4n+3}^{+\frac{1}{2}\pi} &= P_{4n+3}^{+\frac{1}{2}\pi} - P_{4n+2}^{-\frac{3}{2}\pi} = \\ &= -\frac{P_0}{2} \cdot (\delta_{4n+3} + \delta_{4n+2} + \delta_{4n+1} + \delta_{4n+0}) \end{aligned} \tag{15}$$

and can be generalized for every odd bias sample, where it shall be calculated (this results from the demodulation algorithm assumptions [5])

$$e_{2n+1} = -\frac{P_0}{2} \cdot (\underbrace{\delta_{2n+1}}_{\text{current error}} + \underbrace{\delta_{2n+0} + \delta_{2n-1} + \delta_{2n-2}}_{\text{three previous drive errors}}). \tag{16}$$

The feedback signal, used for loop closure, depends on four recent drive values. Because the DAC can be controlled with Δv resolution, the scale factor error, and thus the occurring errors δ_n are all controllable with a fixed delta of $\Delta\phi_v$ corresponding to Δv. In particular, the presented error e_{2n+1} for the last four samples can be controlled with $\Delta\phi_v$ accuracy, but this may only happen when value calculation of the current sample accounts for the three previous ones. These dependencies directly constrain the algorithm design, as discussed in the following section.

VI. Algorithm Design

From the derived error (16), a few important findings can be determined. In this section, the few likely mistakes in algorithm design are first emphasized, followed by a correct modulator implementation presented in pseudocode.

A. Naive implementation

Three pitfalls arise from overlooking details of the derivation in Section V. Each is subtle enough to go unnoticed in operation, yet each degrades performance.

1) Number of samples considered: The modulation algorithm is one of the building blocks of the control loop implemented in the device. Since the scale factor feedback signal is obtained by comparing two consecutive intensity levels (15), an intuitive conclusion follows, that the control loop shall operate at the rate corresponding to the period of two samples. While being correct, this assumption requires additional constraints, otherwise it leads to a naive and incorrect solution. From (16) it is clear, that the feedback signal depends on four adjacent samples, not two. Without the conducted derivation this would not be immediately obvious. A naive algorithm only considering one sample pair could easily lead to control loop oscillations and performance degradation. A correct solution is to always account for the recent two drive values, when calculating the following two.

2) Drive symmetry: The intended drive values between samples n and $n+2$ are symmetrical, i.e. they differ in sign, but have identical amplitude. However, for an actual signal driven with a DAC, this assumption falls short. This is due to the discrete nature of a DAC. In an ideal scenario, a symmetric drive would be desired, but this assumes perfectly controlled amplitude. When the amplitude is discretized, the most reasonable solution is to utilize as much of the resolution, while keeping the drive signals symmetrical as a second priority. This allows for having a $1[lsb]$ amplitude difference, which results in phase shift control accuracy of a single lsb. Otherwise, the effective resolution would be lowered by one bit - amplitude of the effective phase shift would correspond to

$$(k_n \cdot \Delta\phi_v) - (-k_n \cdot \Delta\phi_v) = 2k_n\Delta\phi_v \tag{17}$$

which for adjacent k_n values yields the available accuracy of $2\Delta\phi_v$, while the DAC offers control resolution of a single $\Delta\phi_v$.

3) Rounding factor: Another design corner cut is the rounding method selection for driving discrete DAC values while performing calculations with fractional resolution. Especially prone designs are the ones using floating-point arithmetic (e.g. IEEE-754 float/double, typically used in C/C++ or programmable logic). The problem lies in the rounding direction when working with both positive and negative values (which is the case for driving bias points both ways from zero). Mathematical symmetry of this operation leads to resolution loss. This is regardless of whether the selected rounding algorithm is:

- truncation, or rounding towards zero, e.g. $(-2.2 \rightarrow -2,\ 1 \leftarrow 1.7)$,
- mathematical rounding, e.g. $(-3 \leftarrow 2.9,\ 2.1 \rightarrow 2,\ 1 \leftarrow 1.1, 1.5 \rightarrow 2)$,
- rounding away from zero, e.g. $(-2.4 \rightarrow -3,\ 2.4 \rightarrow 3)$,

TODO check example for all three!!!

If the algorithm rounds fractional values symmetrically, it will lead to the same resolution degradation as described in the previous section. While the effect is the same, the reason is different as the algorithm may not necessarily be implemented to use fractional arithmetics - even though it is the recommended approach. This case is worth emphasizing, as multiple implementations are expected to use floating-point arithmetic in particular. This important numerical issue arises when performing calculations in floating point and simply mapping the result to binary or U2 format required by a DAC. The following snippet shows an example.

```
// intended half-wave voltage in DAC units [lsb]
uint8_t scaleFactor = 14u;

// calculated DAC drive values
int8_t dac[4n+0] = scaleFactor *  0.75 =  10.5 =>  11;
int8_t dac[4n+1] = scaleFactor * -0.25 = -3.5  =>  -4;
int8_t dac[4n+2] = scaleFactor * -0.75 = -10.5 => -11;
int8_t dac[4n+3] = scaleFactor *  0.75 =  3.5  =>   4;
```

Fig. 7: Example of the possible difference between the intended and actual drive values in the context of applying mathematical rounding method.

TODO LABELS CORRECT

In effect the resulting drive does not always correspond to the intended difference, even though it is calculated in $\Delta\phi_v$ domain. In the presented example the result is 30 while the intended scale factor indicates it shall be 28 (effective difference for Equation 16):

$$\begin{aligned} ((4)-(-4))-((-11)-(11)) &= 30 \quad [\Delta\phi_v] \\ ((3.5)-(-3.5))-((-10.5)-(10.5)) &= 28 \quad [\Delta\phi_v] \end{aligned} \tag{18}$$

Additional problem is that this only happens at the level of the least significant bit, so the only reasonable way to track this error is through numerical analysis or simulation of the actual algorithm. It will not be visible in operation, due to the error magnitude relative to the rest of the waveform (Figure 8).

A correct implementation must address all three pitfalls simultaneously, as described next.

B. Optimal implementation

Correct implementation, presented in the snippet in Figure 9, accommodates the mentioned pitfalls. It controls four recent samples at all times and it selects a rounding method,

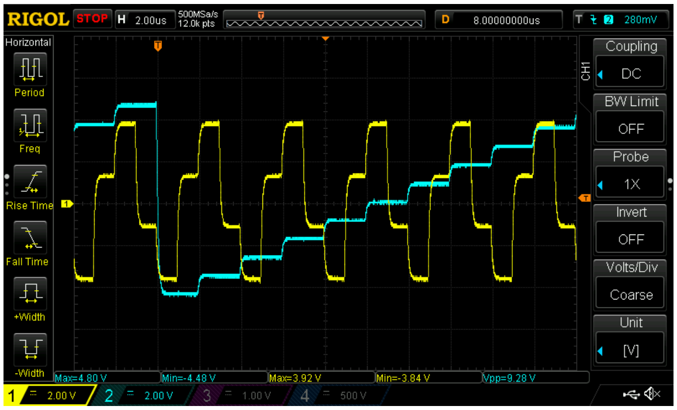


Fig. 8: Modulation pattern acquired using *Sagnac Development Kit* presents an 18b–DAC operating at f=1MHz. Yellow waveform is the biasing sequence, symmetical around zero. Even with a low resolution DAC determining errors at the lsb level would not be possible visually. The cyan waveform in the nulling ramp with Serrodyne reset at 2π.

which has two advantages. Firstly, it is correct in terms of modulating with the accuracy of a single lsb. Secondly, it is implementation–efficient. It utilizes the property of U2 representation, which always rounds values towards $-\inf$, resembling mathematical $floor$ operation. It also uses fixed–point arithmetics to perform calculations with fractional precision. The format used is Q6.2, where the values $\pm\frac{1}{4}$ and $\pm\frac{3}{4}$ are simply expressed as integers ±1 and ±3. For the final result the fractional bits are simply dropped - using right–shift operation in this case. It has to be noted, that fixed–point arithmetics is an optimal choice for implementing an algorithm which does not deal with values of high dynamic range and where determinism regarding accuracy is of paramount importance. Moreover, a floating–point format will also utilize more resources - both in the context of performing operations (addition, multiplication, etc.) and also for value storage (they require additional bits for storing both mantissa and exponent).

```
{ // two previous (already driven) bias points [4n-2, 4n-1]
uint8_t scaleFactor[4n-2] = 13;

int8_t dac[4n-2] = (scaleFactor[4n-2] *  0x3) >> 2 = 9;
int8_t dac[4n-1] = (scaleFactor[4n-2] * -0x1) >> 2 = -4;
}

{ // two current (to be driven) bias points [4n+0, 4n+1]
uint8_t scaleFactorIntended[4n+0] = 14u;
uint8_t scaleFactor[4n+0] =
    (2 * scaleFactorIntended[4n+0]) - scaleFactor[4n-2];

int8_t dac[4n+0] = (scaleFactor[4n+0] * -0x3) >> 2 = -12;
int8_t dac[4n+1] = (scaleFactor[4n+0] *  0x1) >> 2 = 3;
}
```

Fig. 9: Pseudocode for controlling the scale factor for the current two bias points while accounting for the values driven previously. The arithmetics is implemented using Q6.2 fixed–point format. Rounding towards $-\inf$ is natively handled for this format.

In this case the effective drive corresponds to the scale factor, which was intended for the latter two bias points, but it had to account for the previous pair:

$$((3) - (-4)) - ((-12) - (9)) = 28 \quad [\Delta\phi_v] \tag{19}$$

Another possible implementation is to simply execute the control loop once every four samples. This results in very straightforward implementation, which apriori determines all four drive levels and applies them. However, an important drawback of such a solution is that only two of four bias points from Equation 4 are usable for both scale factor and rotation calculation. This results from the dependence of the rotation measurement accuracy on scale factor in each of the bias points. A detailed analysis of this matter remains a subject for further work.

VII. Conclusions and Further Work

An all–digital hardware architecture was presented and advocated for, as it offers the greatest potential for performance improvements through reduced analog complexity and software–defined flexibility. The key finding of the conducted analysis is that the scale factor feedback signal depends on four consecutive drive samples rather than two — a non–obvious result with direct consequences for algorithm correctness. Three implementation pitfalls were identified, and an optimal fixed–point algorithm was derived that recovers one bit of effective resolution through correct sample accounting and rounding. This work requires continuation: the mathematical analysis must also be applied to rotation calculation — in particular, the effect of generating a ramp signal with Serrodyne modulation — intentionally omitted at this time. Furthermore, a complete analysis including the demodulation process and control loop closure needs to be provided to ensure the whole design is well optimized — because the system is as performant as its least performing element.

References

[1] Post, Evert Jan. "Sagnac effect." Reviews of Modern Physics 39.2 (1967): 475.
[2] Sagnac, Georges. "L'éther lumineux démontré par l'effet du vent relatif d'éther dans un interféromètre en rotation uniforme." CR Acad. Sci. 157 (1913): 708-710.
[3] Pavlath, George A. "Closed-loop fiber optic gyros." Fiber Optic Gyros: 20th Anniversary Conference. Vol. 2837. SPIE, 1996.
[4] Babu, G. Harish, A. Venkata Anuhya, and N. Venkatram. "Digital signal processing scheme for open loop and closed loop IFOG using MATLAB/SIMULINK." Indian Journal of Science and Technology 9.11 (2016): 1-10.
[5] Lefevre, Herve C. "Fiber optic gyroscopes." Optical Fiber Sensors. Dordrecht: Springer Netherlands, 1993.
[6] Wang, Wei, Junlong Wang, and Zhengxin Zhao. "Method to control the gain in modulation chain of closed-loop fiber optic gyroscope with periodical biasing modulation." Optical Engineering 51.6 (2012): 064401-064401.
[7] Salvestrini, Jean Paul, et al. "Analysis and control of the dc drift in linbo3-based mach–zehnder modulators." Journal of lightwave technology 29.10 (2011): 1522-1534.
[8] Ulversoy, Tore. "Software defined radio: Challenges and opportunities." IEEE Communications Surveys & Tutorials 12.4 (2010): 531-550.
[9] Liu, Zongwei, Wang Zhang, and Fuquan Zhao. "Impact, challenges and prospect of software-defined vehicles." Automotive Innovation 5.2 (2022): 180-194.